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Final Report

I chose to create a number guessing game for my final project. It is a single player, terminal based game that generates a random number within a range from 1 to 100. The program provides feedback, whether the guess is too high, too low, or correct, after each attempt. If the input is not an integer, it is considered invalid. When the number is guessed correctly, the game ends and displays the number of attempts. I also added a feature to replay the game if desired (without restarting the program) and keep track of the lowest number of attempts to guess the number.

The approach I used was to organize the code with a class and functions. The class handles the details like generating the number, tracking the attempts, and checking the guesses. I use a loop to continue the game until the number is guessed. The program checks if the input is valid and within the range before evaluating the guess. I use another loop to ask the player if they wish to replay the game and check for a valid answer before either starting a new game or ending the game.

For the data structures, I use a list to store the attempt history. This allows for a way to determine the best round. I also include a class to organize the game's logic in a clear way.

One challenge was understanding how to apply data structures into the project. I decided to add the attempt history to incorporate a list, which allowed me to keep count of the best guessing round. Another challenge was handling inputs. As I tested out my code, I realized that I

missed some invalid input cases, so I fixed that by ensuring the input is an integer, adding `.strip()`, and `.lower()` in the `replay` function.

For improvements, I could have added difficulty levels by increasing the number range and limiting the number of attempts allowed. I could have provided more helpful hints by telling the user how close they are to the correct number. These additions could improve user experience and make the game more challenging and fun.